Chapter 4 Heat, light and sound

EXTRA WORKSHEET 1:

Measuring light

SCIENCE UNDERSTANDING **LOGICAL/MATHEMATICAL** **VISUAL/SPATIAL**

Name:_____Class:_____Date: _____

A luminous object produces its own light, whereas a non-luminous object reflects light from another source. For us to see a thing clearly, it must be illuminated by sufficient light. Your eyes are particularly sensitive to yellow-green light (of wavelength 550 nm) and less sensitive to red or blue light. This can be seen in the peak of the curve shown in Figure 1.

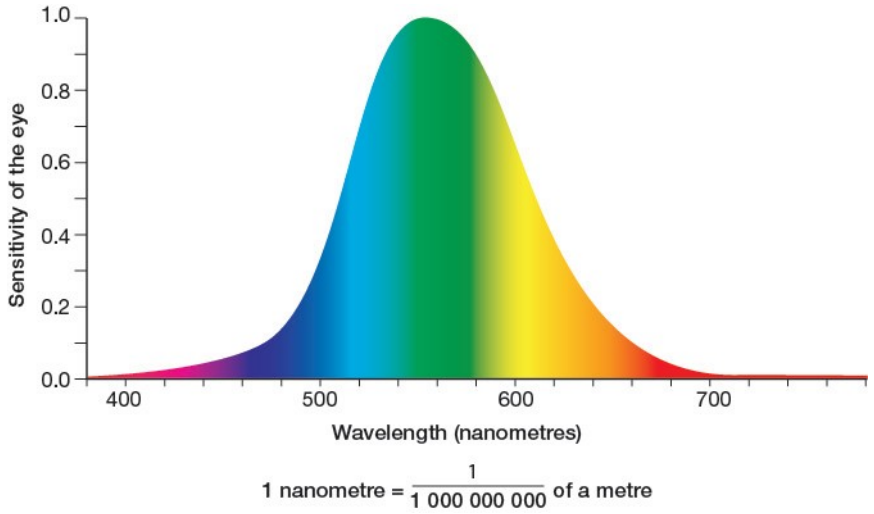


Figure 1 Colour sensitivity of the eye

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This means that if a green light and a red light of equal intensity shine towards you, you will see the green light as being brighter. The luminous intensity of a source, measured in a unit called the candela, takes into account the sensitivity of a typical person's vision to different wavelengths of light. For example, an incandescent 60 W globe appears to have the same luminous intensity as a 9 W compact fluorescent globe. This happens because the incandescent globe emits a lot of heat radiation and less of the colours of the spectrum that the eye is sensitive to compared to the compact fluorescent. (Note that 1 candela is very roughly equal to the light produced by one large candle.)

A light meter measures the illuminance of a source, using a unit called the *lux*. The illuminance of a source is a measure of the intensity of light falling on a surface per unit area. The approximate illuminance (in lux) of a number of sources is shown in the table below.

Illuminance (lux)	Approximate example
0.00005	starry night
1	moonlight
300	office lighting
400	sunrise or sunset
1000	TV studio lighting
1000	overcast day
32 000–100 000	direct sunlight

Questions

1 **State** the unit of measurement of luminous intensity.

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2 **Recall** which frequencies of light the human eye is most sensitive to.

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3 **Explain** why a 9 W compact fluorescent globe appears to have the same brightness as a 60 W incandescent globe.

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4 **a** **Define** the term *illuminance*.

.....

b **State** the unit of measurement of illuminance.

.....

5 **Propose** why measurements of illuminance are important in:

a sporting arenas in which games are played at night

.....

b professional photography

.....

c comparing visual display units such as televisions and computer screens.

.....

6 Study the recommended light levels for the following environments.

Activity	Illuminance (lux)
working in a classroom	250
supermarket shopping	750
working in an operating theatre	1000
drafting architectural designs	1500

Discuss why you think these activities differ in the levels of recommended illuminance.

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